06/12/24 MIS

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[23BSMA31]

THIRD SEMESTER B. TECH EXAMINATION, DECEMBER - 2024

FUNCTIONAL TRANSFORMS & NUMERICAL METHODS

Time: 3 Hours **Maximum Marks: 100**

Instructions: **(i) Missing data may be suitably assumed**

(ii) Answer all Questions

		PART-A			**5 x2=10**
1.	**a.**	Write Dirichlet's conditions for a Fourier series to exist.	L1	CO1	**02**
	b.	Define Fourier sine and cosine transforms.	L1	CO1	**02**
	c.	How do you find the Z-transform of a given function?	L1	CO1	**02**
	d.	Write normal equations for curve $y = ae^{bx}$ using least squares method.	L2	CO1	**02**
	e.	Write the formula for Simpson's 3/8th rule in numerical integration.	L1	CO1	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Find the Fourier expansion of $f(x) = \begin{cases} -k, -\pi < x < 0 \\ k,\ 0 < x < \pi \end{cases}$. Hence deduce that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \ldots$	L3	CO2	**05**
		OR			
	b.	Obtain half range sine series of the function $f(x) = x^2\ in\ 0 < x < \pi$.	L3	CO2	**05**
3.	**a.**	Find the Fourier transform of the function $f(x) = \begin{cases} 1, & \lvert x\rvert \le a \\ 0, & \lvert x\rvert > a \end{cases}$.	L3	CO3	**05**
		OR			
	b.	Obtain the Z-transform of $sinhn\theta$.	L3	CO3	**05**
4.	**a.**	Compute the coefficient of correlation for the following data:	L3	CO5	**05**

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

		OR			
	b.	If θ is the acute angle between the two regression lines relating the variables x and y, show that $tan\theta = \frac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}\left[\frac{1-r^2}{r}\right]$.	L3	CO5	**05**
5.	**a.**	Use Newton Rapson method to find a real root of the equation $xsinx + cosx = 0$ near the point $x = \pi$. Carry out the iterations up to four decimal places of accuracy.	L3	CO6	**05**
		OR			
	b.	Use Regula–Falsi method to find the root of the equation $x^3 - 2x - 5 = 0$ correct to three decimal places.	L3	CO6	**05**
6.	**a.**	Solve the integral equation $\int_0^\infty f(\theta)\cos\alpha\theta\, d\alpha = \begin{cases} 1-\alpha, & 0 \le \alpha \le 1 \\ 0, & \alpha > 1 \end{cases}$	L3	CO3	**05**

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		OR			
	b.	Find the inverse Z-transform of the function $\frac{z}{(z-1)(z-2)}$.	L3	CO3	**05**
7.	**a.**	Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using Simpson's (1/3)rd rule by taking four equal strips.	L3	CO6	**05**
		OR			
	b.	Evaluate $\int_0^1 \frac{dx}{1+x}$ using Simpson's (3/8)th rule by taking seven ordinates.	L3	CO6	**05**
		PART-C			**6 X 10=60**
8.	**a.**	Obtain the Fourier series to represent $f(x) = 2x - x^2$ in $0 \le x \le 2$.	L3	CO2	**10**
		OR			
	b.	Determine the constant term and the first cosine and sine terms of the Fourier series expansion of y from the following table	L3	CO2	**10**

x°	0	45	90	135	180	225	270	315
y	2	3/2	1	½	0	½	1	3/2

9.	**a.**	Find the Fourier transform of $f(x) = \begin{cases} 1-x^2, & \lvert x\rvert < 1 \\ 0, & \lvert x\rvert \ge 1 \end{cases}$. Hence Evaluate $\int_0^\infty \frac{x\cos x - \sin x}{x^3} dx$	L3	CO3	**10**
		OR			
	b.	Find the Z-transform of the following: $(i)\ (\cos\theta + i\sin\theta)^n$ $(ii)\ (n+1)^2$	L3	CO3	**10**
10.	**a.**	If the coefficient of correlation between two variables x and y is 0.5 and the acute angle between their lines of regression is $\tan^{-1}(3/5)$. Show that $\sigma_y = 2\sigma_x$ or $\sigma_x = 2\sigma_y$.	L3	CO5	**10**
		OR			
	b.	Fit a second degree parabola $y = ax^2+bx+c$ in the least square sense for the following data and hence estimate y at $x = 6$.	L3	CO5	**10**

x	1	2	3	4	5
y	10	12	13	16	19

11.	**a.**	From the following data estimate the number of students i) who have scored less than 70 marks ii) between 40 and 45 marks	L3	CO6	**10**

Marks	30-40	40-50	50-60	60-70	70-80
No. of Students	31	42	51	35	31

		OR			
	b.	Use Newton's divided difference formula to find $f(4)$ and $f(7)$ given the data:	L3	CO6	**10**

x	0	2	3	6
$f(x)$	-4	2	14	158

12.	**a.**	Evaluate $\int_0^1 \frac{x}{1+x^2} dx$ using Weddle's rule taking 7 ordinates and hence find $\log 2$.	L3	CO6	**10**

		OR			
	b.	Given $\frac{dy}{dx} = x - y^2$ and the data $y(0) = 0, y(0.2) = 0.02, y(0.4) = 0.0795,$ $y(0.6) = 0.1762,$ Compute y at $x = 0.8$ by applying Milne's predictor corrector method.	L3	CO6	**10**
13.	**a.**	The area of a circle (A) corresponding to diameter (D) is given below: D: 80, 85, 90, 95, 100 A: 5026, 5674, 6362, 7088, 7854 Find the area A for D=105.	L3	CO6	**10**
		OR			
	b.	Use Newton's divided difference formula to find $f(8)$ and $f(15)$ from the following data. x: 4, 5, 7, 10, 11, 13 $f(x)$: 48, 100, 294, 900, 1210, 2028	L3	CO6	**10**